Checkpoint 3 H4

Name: Nazifullah Part #4H4

$C_1=6 \quad C_2=7 \quad a_1=4$

Part (a) optimical solution and slop

$y_1+y_2 = 1+\epsilon$

$y_2+y_3 = 1$

$y_1+y_3 = 1$

$y_2-y_1=0 \Rightarrow y_1=y_2=\frac{1+\epsilon}{2}$

$y_3=1-y_2$

$y(\epsilon)=\left(\frac{1+\epsilon}{2}, \frac{1+\epsilon}{2}, \frac{1-\epsilon}{2}\right)$

$z(1+\epsilon) = 6\frac{1+\epsilon}{2}+7\frac{1+\epsilon}{2}+5\frac{1-\epsilon}{2}$

$= \frac{18+8\epsilon}{2} = \boxed{9+4\epsilon}$

$6y_1+7y_2+5y_3 = 4(y_1+y_2)+3(y_2+y_3)+2(y_1+y_3)$

$\geq 4(1+\epsilon)+3+2 = 9+4\epsilon$

$0 \leq y_j \leq 1$

$z(1) = 9{,}000$

$\boxed{\frac{dz(1+\epsilon)}{d\epsilon}}$

$\boxed{=4{,}000 = a_1}$

✓ Final answer of (A)

b) Valid Range

$0 \leq \frac{1+\epsilon}{2} \leq 1 \Rightarrow 1 \leq \epsilon \leq 1$

$0 \leq \frac{1-\epsilon}{2} \Rightarrow -1 \leq \epsilon \leq 1$

$\boxed{-1 \leq \epsilon \leq 1}$ solution cost

c) with this Range increasing The First Customers Required Caverge by ϵ increase The optimal Cost 4ϵ. So That marginal cost 4 Cost Unit each unit Caverge

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